

FUYUN, China

WMO: 510870

Lat: 46.9941N

Lon: 89.5109E

Elev: 862

StdP: 91.39

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.9	(c) -30.0	(d) -35.9	(e) 0.1	(f) -32.8	(g) -32.9	(h) 0.2	(i) -29.9	(j) 4.5	(k) -13.3	(l) 3.4	(m) -14.4	(n) 0.2	(o) 140	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.7	(c) 33.9	(d) 16.4	(e) 32.2	(f) 15.9	(g) 30.5	(h) 15.5	(i) 18.1	(j) 27.6	(k) 17.3	(l) 27.3	(m) 16.7	(n) 26.8	(o) 3.7	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	12.3	20.0	14.5	11.4	19.9	13.4	10.6	20.0	54.7	26.9	52.3	27.3	50.3	26.9	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 7.8	(c) 6.5	(e) -35.1	(f) 36.9	(g) 4.1	(h) 2.0	(i) -38.1	(j) 38.4	(k) -40.5	(l) 39.5	(m) -42.8	(n) 40.7	(o) -45.8	(p) 42.1		
(a) 9.4	(b) 7.8	(c) 6.5	(e) -35.2	(f) 19.3	(g) 4.1	(h) 1.3	(i) -38.2	(j) 20.2	(k) -40.6	(l) 20.9	(m) -42.9	(n) 21.6	(o) -45.8	(p) 22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-18.2	-14.0	-3.5	8.7	15.2	21.2	23.2	21.3	14.8	6.8	-4.2	-14.8		
	DBStd	15.46	5.71	5.74	7.89	5.44	4.50	3.43	3.04	3.38	4.54	4.96	7.34	5.75		
	HDD10.0	3390	874	672	421	84	9	0	0	0	11	122	427	768		
	HDD18.3	5328	1133	906	677	289	116	13	2	12	121	357	677	1027		
	CDD10.0	1497	0	0	3	46	171	336	411	350	157	24	1	0		
	CDD18.3	393	0	0	0	1	19	99	154	103	17	0	0	0		
	CDH23.3	4464	0	0	0	17	272	1088	1654	1188	240	5	0	0		
	CDH26.7	1736	0	0	0	2	70	414	710	478	61	0	0	0		
(14) Wind	WSAvg	1.8	0.5	0.7	1.6	2.7	2.9	2.9	2.7	2.4	2.1	1.7	1.0	0.7		
(15) Precipitation	PrecAvg	167	8	7	10	14	15	16	23	13	15	14	19	13		
	PrecMax	275	25	19	37	31	50	52	75	32	51	36	45	48		
	PrecMin	83	1	1	1	0	1	1	3	1	2	3	1	1		
	PrecStd	53	7	5	9	8	14	13	19	9	12	8	12	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.0	1.8	17.8	25.5	31.3	34.7	37.5	35.6	31.4	23.8	14.5	1.0		
		MCWB	-3.1	-1.9	6.9	11.3	14.3	16.5	17.9	16.9	14.7	10.4	5.5	-2.1		
	2%	DB	-4.9	-1.1	14.3	22.6	28.6	32.7	34.4	33.2	28.3	20.7	11.0	-2.0		
		MCWB	-6.0	-3.4	5.8	9.9	13.4	15.9	17.0	16.1	13.2	8.8	4.1	-3.3		
	5%	DB	-7.2	-3.4	10.9	20.2	26.4	30.9	32.3	31.2	26.1	18.0	8.0	-4.0		
		MCWB	-8.3	-5.1	4.1	8.7	12.5	15.3	16.4	15.6	12.4	7.8	2.7	-5.3		
	10%	DB	-9.6	-5.3	7.8	18.0	24.2	29.1	30.5	29.5	24.0	15.4	4.9	-6.2		
		MCWB	-10.5	-6.6	3.3	7.9	11.5	15.0	16.0	15.1	11.6	6.9	1.2	-7.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.1	-1.5	8.0	12.3	15.1	18.2	19.6	18.7	15.4	11.1	6.4	-1.3		
		MCDB	-2.3	1.3	15.1	21.5	28.3	27.3	29.4	26.2	27.6	20.6	12.4	0.1		
	2%	WB	-5.9	-3.3	6.3	11.0	14.1	17.3	18.6	17.5	14.3	9.7	4.5	-3.2		
		MCDB	-5.0	-1.2	13.0	19.5	25.7	27.4	28.4	27.9	25.1	17.6	10.0	-2.0		
	5%	WB	-8.2	-5.0	4.8	9.9	13.3	16.6	17.7	16.8	13.3	8.6	3.0	-5.3		
		MCDB	-7.1	-3.5	10.1	18.0	23.9	27.0	27.4	27.4	23.6	16.2	7.3	-4.1		
	10%	WB	-10.6	-6.6	3.3	8.7	12.4	15.9	17.1	16.1	12.4	7.6	1.3	-7.3		
		MCDB	-9.6	-5.3	7.8	16.3	22.0	26.2	26.8	26.2	21.9	14.6	4.6	-6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	13.0	11.7	11.8	12.0	12.1	11.7	12.9	13.4	11.8	10.2	10.4		
		MCDBR	11.0	13.1	12.8	15.1	15.0	14.9	14.6	15.5	16.1	16.0	14.0	10.2		
		MCWBR	9.8	11.1	6.7	6.4	5.3	4.5	4.2	5.0	6.2	7.7	8.0	8.6		
		MCDBR	10.7	12.3	11.6	12.8	13.0	12.1	11.5	12.4	13.7	14.1	12.7	10.0		
	5% WB	MCWBR	9.5	10.4	6.5	5.8	5.0	3.8	3.6	4.1	5.4	7.1	7.5	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.309	0.352	0.397	0.386	0.400	0.393	0.371	0.341	0.331	0.301	0.279		
	taud		2.176	2.103	2.113	2.116	2.219	2.254	2.300	2.337	2.373	2.303	2.244	2.156		
	Ebn at Noon		799	841	858	853	878	865	866	872	869	813	759	737		
	Edh at Noon		84	113	132	146	136	133	125	115	101	92	79	77		
(44) All-Sky Solar Radiation	RadAvg		1.76	2.78	4.16	5.57	6.74	7.11	6.90	6.09	4.87	3.04	1.68	1.30		
	RadStd		0.13	0.15	0.28	0.29	0.38	0.37	0.32	0.20	0.28	0.24	0.21	0.14		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.58	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	-0.40	-0.66	N/A	N/A	N/A	N/A

Nomenclature: See separate page